

SECTION-I
Straight Objective Type

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- The solution of equation $\frac{dy}{dx} = \frac{ax+q}{by+f}$ represents a circle only if
 (1) $a = b$ (2) $a = -b$ (3) $a = -2b$ (4) $a = 2b$
- Which of the following is not a solution of the differential equation $y \frac{dy}{dx} = x \left(\frac{dy}{dx} \right)^2 + 1$
 (1) $y = 4x + \frac{1}{4}$ (2) $3y - 9x = 1$ (3) $y + 2x = 0$ (4) $y = x + 1$
- The solution of differential equation $\frac{dy}{dx} = \frac{y}{2y(\sin y^2 + \cos y^2) - x}$ is
 (1) $xy + \cos y^2 - \sin y^2 = C$ (2) $xy + \cos 2y = C$
 (3) $xy - y^2 = C$ (4) $xy = C$
- The differential equation $y \frac{dy}{dx} + x = a$ where 'a' is a fixed constant represents a set of
 (1) circles having centre on y-axis (2) parabolas
 (3) circles having centre on x-axis (4) straight lines
- The solution of differential equation $2xy \sec^2(y^2) \frac{dy}{dx} + y^4 = \ln(x^2 e^{y^4})$ is
 (1) $\sec^3 y^2 = (\ln x)^2 + C$ (2) $\sec^3 y^2 = 12 \ln x + C$
 (3) $\tan y^2 = (\ln x)^2 + C$ (4) none of these
- The general solution of equation $\frac{dy}{dx} = \frac{3x - 4y + 1}{4x + 3y + 1}$ is
 (1) $(x + 3y)(y - 3x) + 2(y - x) = C$ (2) $(x - 3y)(y + 3x) + 2(y - x) = C$
 (3) $(3y - x)(y + 3x) + 2(y - x) = C$ (4) none of these
- The degree and order of differential equation of the family of all parabolas whose axis is x-axis are respectively
 (1) 2, 1 (2) 1, 2 (3) 1, 1 (4) 2, 3

8. The solution of differential equation $yy' = x \left(\frac{y^2}{x^2} + \frac{f(y^2/x^2)}{f'(y^2/x^2)} \right)$ is
- (1) $f(y^2/x^2) = cx^2$ (2) $x^2f(y^2/x^2) = c^2y^2$ (3) $x^2f(y^2/x^2) = c$ (4) $f(y^2/x^2) = cy/x$
9. The solution of differential equation $\frac{dy}{dx} + \frac{2x}{1+x^2} \cdot y = \frac{1}{(1+x^2)^2}$ is.
- (A) $y(1-x^2) = \tan^{-1} x + c$ (B) $y(1+x^2) = \tan^{-1} x + c$
 (C) $y(1+x^2) = \tan^{-1} x + c$ (D) $y(1-x^2) = \tan^{-1} x + c$
10. The differential equation representing the family of ellipses having foci either on the x-axis or on the y axis, centre at the origin and passing through the point (0, 3) is :
- (1) $xy y' - y^2 + 9 = 0$ (2) $xy y'' + x (y')^2 - y y' = 0$
 (3) $xy y' + y^2 - 9 = 0$ (4) $x + y y'' = 0$
11. The general solution of the differential equation $y dy + \sqrt{1+y^2} dx = 0$ represents a family of
- (1) circles (2) ellipses other than circles
 (3) hyperbolas (4) parabolas
12. The degree of the differential equation which has a solution $y^2 = 4a(x + a^2)$ where 'a' is an arbitrary constant, is
- (1) 2 (2) 3 (3) 4 (4) 1
13. The solution of the differential equation $\frac{ydx + xdy}{ydx - xdy} = \frac{x^2 e^{xy}}{y^4}$ satisfying $y(0) = 1$, is
- (1) $x^3 = 3y^3(-1 + e^{-xy})$ (2) $x^3 = 3y^3(1 - e^{-xy})$
 (3) $x^3 = 3y^3(-1 + e^{xy})$ (4) $x^3 = 3y^3(1 - e^{xy})$
14. The order and the degree of the differential equation of all ellipses with centre at the origin, major axis along x-axis and eccentricity $\frac{\sqrt{3}}{2}$ are, respectively :
- (1) 2, 2 (2) 1, 1 (3) 2, 1 (4) 1, 2
15. The solution of differential equation $ye^{\frac{x}{y}} dx = \left(xe^{\frac{x}{y}} + y^2 \right) dy$ is
- (1) $y = e^{\frac{x}{y}} + C$ (2) $x = e^{\frac{x}{y}} + C$ (3) $y = \ell mx + C$ (4) $y = e^{\frac{y}{x}} + C$

16. The solution of differential equation $\left(xy^2 - e^{x^{\frac{1}{3}}}\right)dx - x^2ydy = 0$ is
- (1) $3y^2 = 2xe^{x^{\frac{1}{3}}} + Cx^2$ (2) $3y^2 = 2x^2e^{x^{\frac{1}{3}}} + Cx^2$
 (3) $y = e^{x^{\frac{1}{3}}} + C$ (4) None of these
17. The general solution of the differential equation $\left[2\sqrt{xy} - x\right]dy + ydx = 0$ is (Here $x, y > 0$)
- (A) $\log x + \sqrt{\frac{y}{x}} = c$ (B) $\log y - \sqrt{\frac{x}{y}} = c$ (C) $\log y + \sqrt{\frac{x}{y}} = c$ (D) None of these
18. Solution of the equation $x^2y - x^3 \frac{dy}{dx} = y^4 \cos x$, when $y(0) = 1$, is -
- (1) $y^3 = 3x^3 \sin x$ (2) $x^3 = 3y^3 \sin x$ (3) $x^3 = y^3 \sin x$ (4) None of these
19. The orthogonal trajectories of the family of curves $a^{n-1}y = x^n$ are given by -
- (1) $x^n + n^2y = \text{const}$ (2) $ny^2 + x^2 = \text{const}$ (3) $n^2x + y^n = \text{const}$ (4) $n^2x - y^n = \text{const}$
20. A solution of the differential equation $\frac{dy}{dx} = \frac{1}{xy[x^2 \sin y^2 + 1]}$ is (C is an arbitrary constant) -
- (1) $x^2 (\cos y^2 - \sin y^2 - 2C e^{-y^2}) = 2$ (2) $y^2 (\cos x^2 - (\sin y^2 - 2C e^{-y^2})) = 2$
 (3) $x^2 (\cos y^2 - \sin y^2 - e^{-y^2}) = 4$ (4) None of these

SECTION-II

(Integer Type Questions)

This section contains **Fifteen (15)** questions. The answer to each question is **NUMERICAL VALUE** with two digit integer and decimal upto one digit

21. The order of differential equation of family of curves $Ax + By + C = 0$ where A, B, C are arbitrary constants, is
22. Consider the following differential equations :
- (i) $(x-y)\frac{dy}{dx} = 2x + y$ (ii) $x \cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y \cos\left(\frac{y}{x}\right) + 4x$
 (iii) $2x^2y^2 \frac{dy}{dx} = x^2 + y^2$ (iii) $\sin x \frac{dy}{dx} = \cos x$

How many of these are homogeneous differential equations?

23. Solution of differential equation $\frac{dy}{dx} + \frac{y}{x} = x^2y^6$ is of form $\frac{2}{x^5y^5} = \frac{A}{x^2} + C$. Then find value of A
24. If the solution of the differential equation $\frac{dy}{dx} = \frac{ax+3}{2y+f}$ represents a circle, then the value of 'a' is –
25. Degree of differential equation $e^{dy/dx} = x$ is -

ANSWER KEY_DPP-18

1.	(2)	2.	(3)	3.	(1)	4.	(3)	5.	(3)	6.	(3)	7.	(2)
8.	(1)	9.	(2)	10.	(1)	11.	(3)	12.	(2)	13.	(2)	14.	(2)
15.	(1)	16.	(2)	17.	(3)	18.	(2)	19.	(2)	20.	(1)	21.	(2)
22.	(2)	23.	(5)	24.	(-2)	25.	(1)						